CHANDIGARH UNIVERSITY

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Experiment 04

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Semester	: Sixth	**Date of exp.**	: 03/02/26
Subject	: System Design	**Subject Code**	: 23CSH-314

AIM

To design a scalable OTT platform like Netflix or Amazon Prime.

OBJECTIVE:

- To enable secure user registration and login for personalized access.
- To provide video streaming services with high quality and minimal latency.
- To manage content storage and categorization for movies, shows, and live events.
- To deliver personalized recommendations based on user preferences and history.
- To support subscription plans and payment integration for monetization.
- To ensure scalability, reliability, and data security for millions of concurrent users.

REQUIREMENTS

Functional Requirements:

- Client should be able to create account on the OTT platform.
- After the successfull login, client should be able to opt for the subscription plans.
- Client should be able to search for the shows/movies based on the video title or names.- {Elastic Search}
- Client should be able to watch the videos / tv shows in multiple different resolutions (480p,720p,1080p, 4k etc.)
- Recommendation for TV shows and movies.(Optional)

Non-Functional requirements:

- Scalability: 200-300M, for which let's say total videos we are having are 20K videos (~1 hour each)
- CAP Theorem: Availability >>>>> Consistency

  Availability on watching TV shows and movies

Consistency in making payments and in subscription plans

- Latency: 50 - 80 ms
  Client should be able to see the video with zero or neglibile buffering.

HIGH LEVEL DESIGN

Core-Entities

- Clients
- Clients Metadata
- Video / TV shows
- Video metadata: (images(Thumbnails + description)

Attached ott.drawio file has whole high-level design.

LOW LEVEL DESIGN

API-Endpoints

- Client-Onboarding
  i. POST Call: https://www.netlfix.com/user/register
  ii. POST Call: https://www.netlfix.com/user/login
  iii. PUT Call: https://www.netlfix.com/user/update

- Subscription
  i. GET Call: https://www.netlfix.com/Get-subscription-plans
  ii. POST Call: https://www.netlfix.com/subscription
    1. { userMetadata,
    2. subscriptionID}

- Searching & Video Playing
  i. GET Call: https://www.netlfix.com/search?q={movie_name}
     Response: List<Video_ID> + some meta data of video
     Pagination
  ii. GET Call: https://www.netlfix.com/{video_ID}
     Response: Metadata of the video (JSON)
  iii. GET Call: https://www.netlfix.com/play/{video_ID}

LEARNING OUTCOMES

- I learned how to design an OTT platform using different databases (relational and NoSQL) to manage user data, subscriptions, and content metadata.
- I got to know how Kafka can be used for event streaming to handle real-time notifications and activity logs.
- I learned how to integrate a third-party payment service for secure subscription and transaction management.
- I understood the use of the Proxy Design Pattern to manage requests efficiently and improve system scalability.
- I got to know how Blob Storage and a Chunker are used to store and deliver large video files in smaller segments.
- I learned how video encoders and manifest files enable adaptive streaming and smooth playback across different devices.